CLASS 10 MATHEMATICS :: FA-2

Generating New Problems

Student Name: ..... **Roll No:** ..... **Class & Sec:** .....

1. Check the consistency of the pair of linear equations: $x + 2y = 4$ and $2x + 4y = 12$ using coefficient ratios.

2. Solve the pair of linear equations by the elimination method: $2x + y = 7$ and $3x - 2y = 7$.

3. Find the value of k for which the system of equations $2x + ky = 1$ and $3x - 5y = 7$ has a unique solution.

4. Find the roots of the quadratic equation $x^2 - 7x + 12 = 0$ using the method of factorisation.

5. Calculate the discriminant of $2x^2 - 4x + 3 = 0$ and determine the nature of its roots.

6. Determine the 15th term of the Arithmetic Progression: 3, 8, 13, 18, ...

7. Calculate the sum of the first 20 terms of the AP: 2, 7, 12, 17, ...

8. Which term of the Arithmetic Progression 18, 15, 12, ... is equal to -30?

9. Find the Mode of the following set of raw marks: 4, 7, 5, 7, 6, 7, 8, 5, 9, 7.

10. State the formula used to calculate the Median of a grouped frequency distribution and identify all variables involved.